ClassNest Assessment - Student Question Paper

Class: Python Basics | Unit 1: Functions

Assessment: Unit 1 - Functions Assessment

Total: 46 marks | Suggested Duration: 45 minutes

Instructions	1. Answer all questions. 2. MCQs carry 1 mark each. 3. Fill-ups carry 1 mark each. 4. Coding questions carry the marks mentioned. 5. Coding answers may be manually evaluated by the teacher.

Part A - MCQs (10 x 1 = 10 marks)

1. A subroutine is best described as:

   A. A small section of code used to perform a specific task

   B. A hardware device used for input

   C. A type of database

   D. A file extension

2. In programming languages, a subroutine is also commonly called a:

   A. Variable

   B. Function

   C. Compiler

   D. Loop

3. Which of the following is the main advantage of using functions?

   A. They make repeated code reusable

   B. They remove the need for input

   C. They make code run without memory

   D. They replace all variables

4. In a function definition, the variables listed inside the function header are called:

   A. Arguments

   B. Parameters

   C. Keywords

   D. Constants

5. The actual value passed to a function while calling it is called:

   A. Parameter

   B. Argument

   C. Return type

   D. Module

6. In the specification 'requires: x > 0', the word 'requires' indicates a:

   A. Postcondition

   B. Precondition

   C. Loop condition

   D. Error message

7. In the specification 'returns: square of x', the word 'returns' indicates a:

A. Precondition

B. Postcondition

C. Parameter

D. Recursive call

8. A function that calls itself is called a:

A. Static function

B. Recursive function

C. Selector function

D. Constructor

9. Which statement is true about a recursive factorial function?

A. It must have a stopping/base condition

B. It cannot use parameters

C. It cannot return a value

D. It never terminates by design

10. In the expression square(5), the value 5 is the:

A. Function name

B. Parameter

C. Argument

D. Keyword

Part B - Fill in the blanks (8 x 1 = 8 marks)

1. A small section of code used to perform a specific task is called a __________.

2. The variables in a function definition are called __________.

3. The values passed to a function call are called __________.

4. A condition that must be true before a function executes is called a __________.

5. The condition/result guaranteed after function execution is called a __________.

6. A function definition that calls itself is called a __________ function.

7. In Python, the keyword used to define a function is __________.

8. The statement used to send a value back from a function is __________.

Part C - Simple Coding Questions (28 marks)

1. Write a function square(x) that returns the square of x. **(5 marks)**

Starter code:

```
def square(x):
# write your code here
pass
```

Visible test cases:

```
Input: square(5)
Expected: 25

Input: square(12)
Expected: 144
```

2. Write a function add(a, b) that returns the sum of two numbers. **(5 marks)**

Starter code:

```
def add(a, b):
# write your code here
pass
```

Visible test cases:

```
Input: add(10, 20)
Expected: 30

Input: add(-2, 8)
Expected: 6
```

3. Write a recursive function factorial(n) that returns the factorial of n. Assume n >= 0. **(8 marks)**

Starter code:

```
def factorial(n):
# write your code here
pass
```

Visible test cases:

```
Input: factorial(0)
Expected: 1

Input: factorial(5)
Expected: 120
```

4. Write a function is_even(n) that returns True if n is even, otherwise False. **(5 marks)**

Starter code:

```
def is_even(n):
# write your code here
pass
```

Visible test cases:

```
Input: is_even(4)
Expected: True

Input: is_even(7)
Expected: False
```

5. Write a function max_of_two(a, b) that returns the greater of two numbers. **(5 marks)**

Starter code:

```
def max_of_two(a, b):
# write your code here
pass
```

Visible test cases:

```
Input: max_of_two(8, 3)
Expected: 8

Input: max_of_two(4, 9)
Expected: 9
```